

2025-2026 Hawk & Hornet Invitational

Designer Genes C
EXAM BOOKLET

**DO NOT WRITE
ON THESE PAGES!**



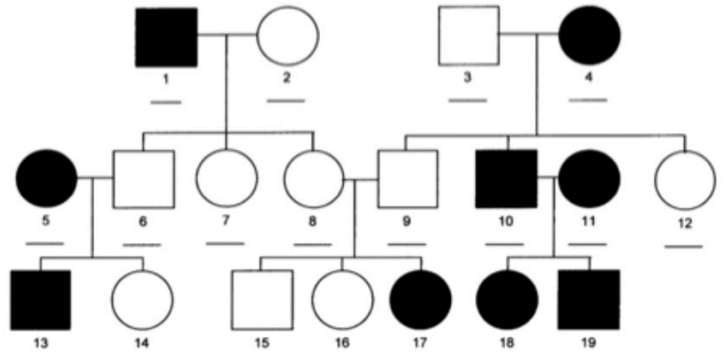
Total Score: ____ / 118

Instructions & Information

- 50 minutes will be given for the exam
- Each team may bring writing utensils, two Class II calculators, and one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
- Tiebreakers will be denoted with **[TB#]** and will be in the following order: 9, 14, 22, 34, 49, 57, 66

Good luck!

1. The chromosomal basis underlying Mendel's Law of Independent Assortment is based on meiosis events occurring in:
 - a. Metaphase I
 - b. Metaphase II
 - c. Anaphase I
 - d. Anaphase II
2. A diploid germ cell with $n = 16$ chromosomes is undergoing meiosis. At the end of telophase of meiosis I, this cell will have split into:
 - a. Two $2x$, $1n$ cells with $n = 16$
 - b. Four $1x$, $2n$ cells with $n = 16$
 - c. Two $1x$, $1n$ cells with $n = 32$
 - d. Four $3x$, $2n$ cells with $n = 32$
3. Some cats exhibiting polydactyly can have up to 28 toes while their identical twins can have a normal number of toes. This inheritance pattern can be best described as:
 - a. Incomplete penetrance
 - b. Haplosufficiency
 - c. Variable expressivity
 - d. Dominant negative
4. Albinism is only observable in individuals who are homozygous for the mutant tyr allele. Given this, the wild type Tyr allele can be described as:
 - a. Dominant to mutant tyr
 - b. Recessive to mutant tyr
 - c. Haplosufficient
 - d. A and C
5. Given a germ cell with the genotype $A/a;B/b$ undergoing meiosis, which of the following are possible genotypes for the gametes formed from this event?
 - a. Aa, Aa, Bb, Bb
 - b. A, a, B, b
 - c. Ab, Ab, aB, aB
 - d. aa, AA, BB, bb
6. Sometimes, more than one chloroplast or mitochondrial genotype can be found in a single organism. This phenomenon is known as:
 - a. Heteroplasmy
 - b. Homoplasmy
 - c. Hemizygy
 - d. Heterozygy
7. Chromosomal DNA is wrapped around proteins known as _____.
 - a. Chromatin
 - b. Chaperones
 - c. Nucleosomes
 - d. Histones
8. You breed a male green beetle with a female green beetle and find that all the offspring they produce are blue. Which of the following is a plausible explanation for this resulting phenotype?
 - a. Maternal effect
 - b. Maternal inheritance
 - c. Autosomal dominant mutation(s)
 - d. All of the above
9. **[TB#1]** Assuming Chargaff's rules apply, what percentage of an organism's genome would be composed of adenine nucleotides if it is given that their genome is 18% cytosine?
 - a. 18%
 - b. 36%
 - c. 32%
 - d. 64%
10. If an organism has 10 chromosomes in prophase II of meiosis, how many chromosomes would it have in prophase of mitosis?
 - a. 5
 - b. 10
 - c. 20
 - d. 40



11. The inheritance pattern shown above is best described as:
 - a. Autosomal dominant
 - b. Autosomal recessive
 - c. Mitochondrial
 - d. X-linked dominant
12. In the cross $AABbCCDdEe \times AaBBCCDdEe$, what proportion of offspring will express the dominant phenotype for all five traits?
 - a. 0.03
 - b. 0.25
 - c. 0.38
 - d. 0.56

13. Which of the following structures does RNA polymerase bind to?
- The promoter on DNA
 - The promoter on RNA
 - The start codon on DNA
 - The start codon on RNA

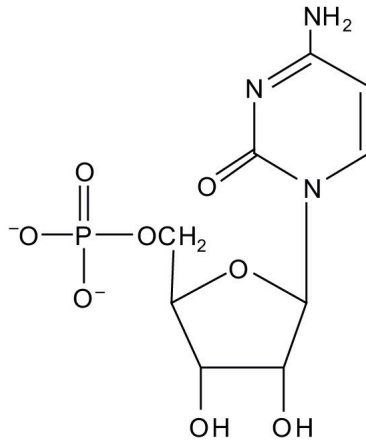
14. **[TB#2]** A deletion of two nucleotides in the middle of an intron will most likely result in a(n):
- Nonsense mutation
 - Silent mutation
 - Missense mutation
 - Frameshift mutation

15. The structure shown to the right is best described as a:

- Nucleotide
- Nucleoside
- Nucleobase
- Nucleosome

16. What is the identity of the previous structure?

- Adenine
- Thymine
- Guanine
- Cytosine



19. A gene knockdown involves targeting _____ sequences of interest.
- DNA
 - mRNA
 - rRNA
 - tRNA

20. Pyrimidine dimer lesions are generally caused by which process?
- Transposons
 - Replication errors
 - UV radiation
 - Spontaneous mutations

21. Which of the following repair pathways is typically considered the most error prone?
- Base excision repair
 - Nucleotide excision repair
 - Non-homologous end joining
 - Mismatch repair

Match the following descriptions with the PCR step– (A) denaturation, (B) annealing, and (C) extension– with which they best correspond.

22. **[TB#3]** _____ Involves extension by Taq or another heat-resistant polymerase such as Pfu.
23. _____ Occurs at roughly 3–5°C lower than the lowest melting temperature (T_m) of primers used.
24. _____ Involves increasing solution temperature to separate double-stranded DNA into single strands.
25. _____ Involves binding of primers to regions which flank the target DNA.
26. _____ Increasing the temperature of this step increases primer specificity.
27. _____ Occurs at roughly 94–98°C.

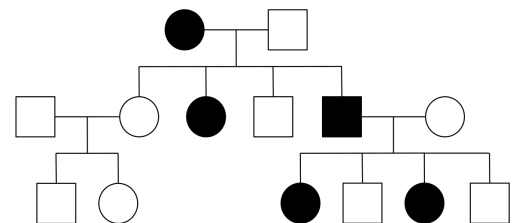
You know that the gene which makes pumpkins orange is dominant to the gene which makes pumpkins white. Both genes are located at the same locus.

Visiting your parents' garden, you find a population of pumpkins which are all orange but you do not know if they are truly true breeding.

You then cross an orange pumpkin from your parents' garden with white, true-breeding pumpkin seeds you've obtained from a botany store. Of the 100 seeds you obtain from your cross, 45 grow orange pumpkins, and the rest grow white pumpkins.

17. Assuming normal Mendelian inheritance, the orange pumpkin you used for the cross was most likely:
- Homozygous
 - Heterozygous
 - Indeterminate

18. The experiment you conducted is best described as a(n):
- P intercross
 - F1 intercross
 - Backcross
 - Testcross

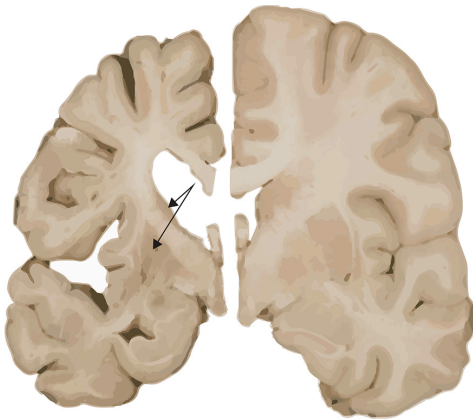


28. Identify the mode of inheritance shown above.
- Maternal
 - Autosomal dominant
 - X-linked dominant
 - Y-linked recessive

You cross 40 violet-flowered garden peas (*Pisum sativum*) to white-flowered garden peas. Violet is dominant to white. Each cross results in 200 seeds. After planting seeds from all your crosses, you find that you have a total of 5,200 plants with violet flowers.

29. True (A) / False (B): In this scenario, all of your white-flowered garden peas must be homozygous for flower color.
30. When crossing a heterozygous violet-flowered pea plant with a white-flowered pea plant, what percentage of resulting offspring would you expect to have white flowers?
- 25%
 - 50%
 - 75%
 - Cannot determine based on the given information
31. How many total seeds will you have planted?
- 200
 - 4,000
 - 5,200
 - 8,000
32. How many of your original violet-flowered pea plants were heterozygous?
- 14
 - 26
 - 28
 - 36
33. How many of your original violet-flowered pea plants were homozygous?
- 4
 - 12
 - 14
 - 26

Huntington's disease is inherited as an autosomal dominant trait. A woman and man both with Huntington's disease each have a parent with the disease, a parent without, and a sibling without.



Huntington Disease Brain

Normal Brain

Courtesy of Dr. Jean Paul Vonsatte
Columbia University

34. **[TB#4]** What is the genotype of the woman's sibling?
- Hh
 - HH
 - hh
 - Hh/hH
35. What is the probability that this couple has three children in a row with Huntington's disease?
- 0.0156
 - 0.0938
 - 0.1250
 - 0.4219
36. What is the probability that the first two children this couple has are a boy with the disease and a girl without, or two girls without the disease?
- 0.0469
 - 0.0625
 - 0.1250
 - 0.1406

Match the following laws of inheritance to their respective descriptors.

37. **A** _____ Law of Dominance. A. In a heterozygote, one trait will conceal the presence of another trait for the same characteristic.
38. **C** _____ Law of Independent Assortment. B. During meiosis, the two alleles for one parental trait are split so each gamete receives only one.
39. **B** _____ Law of Segregation. C. The inheritance of an allele for one trait does not influence the inheritance of an allele for a different trait.

You are conducting a diploid trihybrid cross with the following Plants A and B.

Plant A: $x^0x^1;y^0y^1;z^0z^1$

Plant B: $x^0x^1;y^0y^1;z^0z^1$

Each of the alleles shown contribute to the height of the plant. $x^1=y^1=z^1$ and $x^0=y^0=z^0$.

- | | |
|---|--|
| <p>40. In this model, height is best described as:</p> <ol style="list-style-type: none"> Pleiotropic only Epistatic only Additive and polygenetic Additive and epistatic <p>41. Assuming independent assortment, how many different gametes can be produced by Plant A?</p> <ol style="list-style-type: none"> 6 8 36 64 | <p>42. How many different resulting genotypes will result in a phenotypic value of 4?</p> <ol style="list-style-type: none"> 4 6 9 15 <p>43. What is the frequency of the phenotypic class with a value of 4?</p> <ol style="list-style-type: none"> 0.0625 0.0938 0.1406 0.2344 |
|---|--|

44. True (A) / False (B): Quantitative traits are generally attributed to only one single gene.
45. True (A) / False (B): Stabilizing selection results in increased genetic diversity.
46. True (A) / False (B): Directional selection can result in changes in allele frequency.
47. True (A) / False (B): A recombination frequency of less than 50% at a certain loci in a population indicates that some degree of genetic linkage is present.
48. True (A) / False (B): Eukaryotic ribosomes consist of a small 30S subunit and a large 50S subunit.
49. **[TB#5]** True (A) / False (B): Euchromatin is considered transcriptionally active.

Match the following functions with the respective polymerase with which they are associated. Some polymerases may be used more than once, while some may not be used at all.

A. DNA polymerase I

B. DNA polymerase II

C. DNA polymerase III

D. DNA polymerase δ (delta)

E. DNA polymerase ϵ (epsilon)

AB. DNA polymerase α (alpha)

- | | |
|---|---|
| <p>50. _____ Is involved in leading strand synthesis in eukaryotes.</p> <p>51. _____ Is involved in Okazaki fragment synthesis in eukaryotes.</p> <p>52. _____ The primary enzyme responsible for replication elongation at the 3' end, synthesis, and elongation in prokaryotes.</p> <p>53. _____ Responsible for synthesizing Okazaki fragments in prokaryotes.</p> | <p>54. _____ Present in prokaryotes. This protein has unique 5' to 3' exonuclease activity which allows it to assist in DNA repair, primer removal, and gap filling.</p> <p>55. _____ Has a similar function to DNA polymerase III in prokaryotes and largely serves as a backup in repair and replication.</p> <p>56. _____ Found in eukaryotes, this enzyme has primase activity and initiates DNA synthesis.</p> |
|---|---|

A large number of squirrels on the University of Michigan campus have been genotyped at the Uom locus by student researchers. Their findings are summarized in the table below.

Genotype	Number of squirrels
UomA / UomA	420
UomA / UomB	190
UomB / UomB	260
UomC / UomC	20
UomC / UomD	15



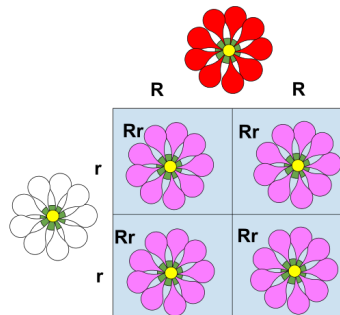
Complementary image of the 2024 UM Fat Squirrel Photo Contest winner by user eekobroom

57. **[TB#6]** What is the frequency of the UomC allele?
- 0.008
 - 0.011
 - 0.019
 - 0.030
58. What is the frequency of the UomB allele?
- 0.105
 - 0.144
 - 0.249
 - 0.392
59. How many UomA alleles are present?
- 400
 - 250
 - 650
 - 1050
60. If this population is in Hardy-Weinberg Equilibrium, what is the expected frequency of the genotype UomA/UomB in the next generation?
- 0.109
 - 0.138
 - 0.481
 - 0.750
61. Which of the following is not an assumption necessary to apply Hardy-Weinberg equilibrium?
- No mutations are present
 - No migration or emigration from the population
 - Nonrandom mating
 - Large population size

Identify whether Illumina sequencing (A) or Nanopore sequencing (B) is most appropriate for the following scenarios.

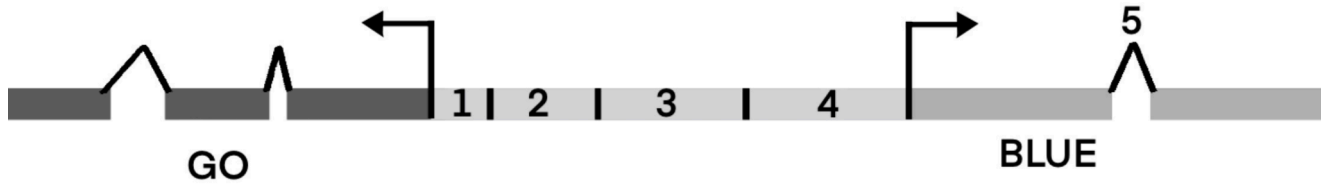
62. _____ Long transcripts (> 1000 kb).
63. _____ Error rate must be minimized.
64. _____ A lab has existing infrastructure which allows for detection of electric current variation associated with specific nucleobases.

65. Identify the method of inheritance shown.
- Incomplete dominance
 - Codominance
 - Epistasis
 - More than one of the above is possible



66. **[TB#7]** The 9:3:3:1 phenotypic ratio is indicative of a:
- Heterozygous monohybrid cross
 - Heterozygous dihybrid cross
 - Heterozygous trihybrid cross
 - Trihybrid test cross

You are characterizing the cis-regulatory regions of genes *GO* and *BLUE* in the wolverine genome. For your analysis, you make mutations in several regions shown below and note their effects on *GO* and *BLUE*.



Mutated Region	Effect on <i>GO</i>	Effect on <i>BLUE</i>
1	Wild-type	No expression
2	Decreased expression	Increased expression
3	Decreased expression	Wild-type
4	No expression	Wild-type
5	Decreased expression in low temperatures	Wild-type

67. [2 pts] Identify the region(s) above which are promoters.

68. [2 pts] What is the most likely function of region 2?

A true-breeding fly with no wings due to a mutation in *wingless* (*wg*) is crossed to a mutant yellow-bodied fly with a mutation in the *yellow* (*y*) gene. Both *wg* and *y* are on the same autosome. All F1 progeny have wings and wild-type brown body color.

69. [4 pts] Assuming recombination occurs during the cross described, write the four haploid gametes produced by an F1 female. Identify which gametes are parental and which are recombinant with respect to the F1 female.

70. [4 pts] If a single nondisjunction event occurs in meiosis II of the cross producing the F1 generation, identify the two possible genotypes of the **abnormal** gametes produced by this cross.

71. [3 pts] Briefly differentiate between maternal effect and maternal inheritance.

72. [2 pts] Identify all components of the mitotic cycle in chronological order.

73. [2 pts] Identify at least one phase present in the cell cycle but NOT in the mitotic cycle.

74. [2 pts] In order, list the three steps of transcription.

Label the following components of the replication fork shown.

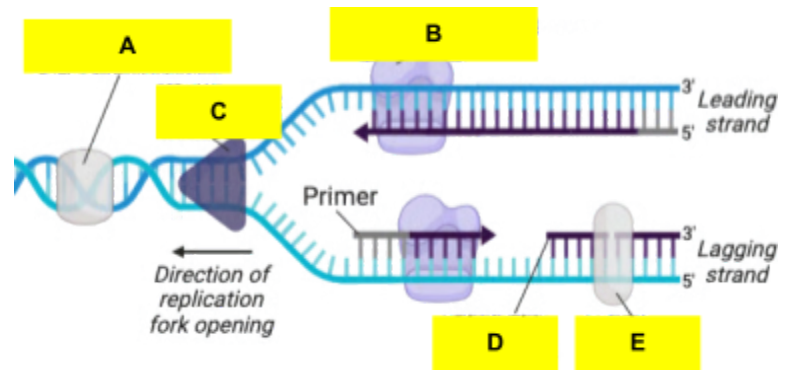
75. _____

76. _____

77. _____

78. _____

79. _____



80. [4 pts] Ten wild-type female zebrafish are crossed to ten male zebrafish which lack tails. This tailless phenotype is caused by a recessive mutation in the *no tail* (*ntl*) gene. Each female zebrafish only mates once and gives birth to 100 baby fish. Of the 1,000 total offspring, 200 lacked tails and 800 had tails. Given this, how many of the original female zebrafish were heterozygous?

Blue-white screening is a molecular biology technique used to identify recombinant bacterial colonies. In *E. coli*, the *lac* operon plays a central role in this process. When a gene of interest is inserted into the *lacZ* gene of a plasmid, it disrupts the production of β -galactosidase, an enzyme that normally cleaves X-gal, a substrate. *E. coli* cells are plated on agar containing X-gal and IPTG (an inducer of the *lac* operon).

- Blue colonies: These arise from bacteria where the *lacZ* gene is intact. The β -galactosidase enzyme is functional, cleaving X-gal and producing a blue color.
- White colonies: These result from bacteria where the *lacZ* gene is disrupted due to the insertion of the foreign gene. Without functional β -galactosidase, no blue color is produced.

Thus, white colonies indicate successful cloning (gene insertion), while blue colonies suggest unsuccessful cloning. This makes blue-white screening a simple and effective method to identify recombinant *E. coli* strains.

81. True (A) / False (B): the Lac operon is considered a repressible operon.

Match the corresponding Lac operon responses of (A) no transcription (B) reduced transcription (C) normal/high transcription to the following scenarios:

82. _____ High glucose and low lactose conditions. 84. _____ cAMP is high in the cell as is lactose.
83. _____ Low glucose and high lactose conditions. 85. _____ cAMP is low and the Lac repressor is bound.

86. According to the description provided of blue-white screening, non-recombinant bacteria will appear as _____ as opposed to _____ colonies which are associated with recombinant bacteria.

- Blue; white
- White; blue
- A mix of blue and white; white
- A mix of blue and white; blue

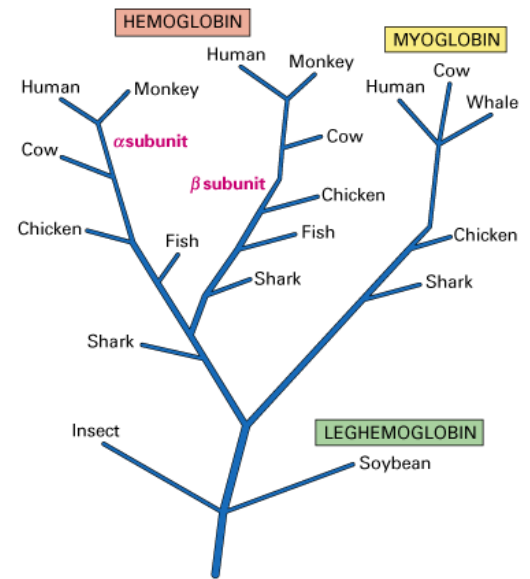
87. [4 pts] If you were attempting to insert a gene of interest into *E. coli* with this blue-white screening method, what color would colonies be which are successfully expressing your gene of interest? Briefly explain.

88. [4 pts] You accidentally plate your *E. coli* on agar which does not contain IPTG. Briefly explain the results you would expect, and why you would expect such.

“Of the remaining functional DNA, 25-50 percent of the genes encoding proteins are found only once in the haploid genome and are known as solitary genes. Frequently, the DNA surrounding a particular gene contains sequences that are close, but not identical, copies of the gene. These multiple copies are believed to be the result of duplication and divergence. This is the process by which a single gene is first duplicated and then undergoes selective pressure to mutate into a gene that is similar, but not identical, in sequence to its ancestral gene. “

“One such example of this process is the development of the beta-globin gene family. After globin family genes from different species were sequenced, an evolutionary tree predicting the development of the globin family of genes was developed. A proposed tree is shown.”

The above excerpt is sourced from Muhlenberg College, The B-globin Gene Family.



89. [2 pts] On the diagram provided on your answer sheet, circle ONE region where gene duplication has occurred.
90. [2 pts] On the diagram provided on your answer sheet, draw two rectangles around TWO genes which are paralogs.
91. [2 pts] On the diagram provided on your answer sheet, draw stars next to TWO genes which are paralogs (same as above, different symbol to distinguish).
92. [2 pts] List two genes which are orthologs.